Math 55 quiz 03/31/2026

You must type or neatly handwrite each answer on a separate piece of paper and select the pages for each question when you upload to Gradescope. Feel free to use more than one sheet for an answer if needed. Note that both PDF and picture/photo format (JPG) are accepted. You must show your work. During the quiz, you may consult the textbook, your own notes and other course materials you yourself prepared before the beginning of the quiz. You may not consult resources online. You can post a private question to Ed Discussion to instructors only, but do not make public posts on Ed Discussion, Discord, or other online forums.

By submitting this quiz, you agree to adhere to the statement below: "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. All of the work submitted here is mine, and I have not discussed the content of this exam with any other person. I have not consulted any resources beyond the course materials and my notes during the exam. I have not used calculators, computers, or the internet to do calculations or look up answers."

1. How many possible (unordered) hands of 5 cards can you make from a 52 card deck?

Solution: This is the number of combinations of 5 elements from a 52 element set:

$$\binom{52}{5} = \frac{52!}{5! \cdot (52-5)!} = 2598960$$

2. How many positive integers below 1 million have no factors of 2, 3, or 5?

Solution: Note that a number n has no factors of 2, 3, or 5 iff n is relatively prime to 30. The numbers in $\mathbb{Z}/30\mathbb{Z}$ relatively prime to 30 are $1, 7, 11, 13, 17, 19, 23, 29$, so there are eight of them. This means that there are $8 \cdot \lfloor \frac{1000000}{30} \rfloor = 266664$ such numbers below 999990. There are two more below 1 million: 999991 and 999997. Hence, there are 266666 such numbers.